

5F OFFICE 85-252 ORDER NO.	5D100114001 WDID 144572 REG MEASURE ID	FACILITIES INSPECTION REPORT	N15 PROGRAM 37493 PARTY ID	1/10 PAGE NO. 273185 PLACE ID
Riverdale Public Utility District DISCHARGER NAME		Riverdale Wastewater Treatment Facility FACILITY NAME		
P.O. Box 248 STREET ADDRESS		APN 053-090-1100 STREET ADDRESS		
Riverdale, CA 93656 CITY, STATE, ZIP CODE		Riverdale, CA 93656 CITY, STATE, ZIP CODE		
Vincent Romero DISCHARGER CONTACT PERSON		Vincent Romero FACILITY CONTACT PERSON		
559-867-3838 TELEPHONE NO.	vromero@riverdale.com E-MAIL ADDRESS	559-867-3838 TELEPHONE NO.	vromero@riverdale.com E-MAIL ADDRESS	

GENERAL INSPECTION INFORMATION

Inspection Type: Compliance B - Pre WDR Lead Inspector: Jeff Pyle

3/07/17 to 3/07/17 2:30 pm Sunny, clear, upper 50's to low 60's. Slight breeze.

INSPECTION DATE(S) INSPECTION TIME GENERAL WEATHER CONDITIONS

INSPECTION ATTENDEE(S) (Identify additional attendees in GENERAL OBSERVATIONS Section)

Vincent Romero	Riverdale Public Utility District	559-867-3838	vromero@riverdale.com
Tristan Mauger	Riverdale Public Utility District	559-867-3838	Tristan.mauger@sbcglobal.net
Maija Madec	Provost & Pritchard	559-326-1100	mmadec@ppeng.com
Matt Kemp	Provost & Pritchard	559-326-1100	mkemp@ppeng.com
NAME	COMPANY/AGENCY	TELEPHONE NO.	E-MAIL ADDRESS

INSPECTION SUMMARY (for CIWQS entry)

Pre Waste Discharge Requirements (WDRs) inspection of the wastewater treatment facility (WWTF) at the Riverdale Public Utilities District (District) WWTF prior to replacement of the old WWTF. Site is old and in need of repair (pond berms). Secondary-treated effluent is discharged into unlined evaporation/percolation ponds, before being discharged to an adjacent ~32-acre land application area. The existing flow being about 85 percent of the designed capacity, standing water in the land application area, and effluent biochemical oxygen demand (BOD) in excess of daily average/maximum limits have driven the proposed upgrade of the WWTF.

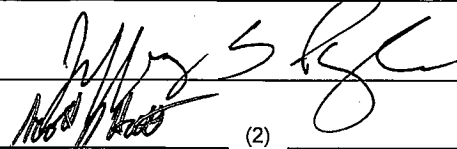
INSPECTION VIOLATIONS SUMMARY

Identify VIOLATIONS noted during inspection in table below. For each violation documented entered into CIWQS, identify Violation ID and Violation Type, describe violation, and identify section of the WDRs or Water Code violated. **None**

Label	Violation ID	Violation Type	Violation Description	Section of the WDRs Violated

OTHER VIOLATIONS

SMR violations?	☒ Yes ☐ No ☐ Not Evaluated	See SELF-MONITORING REPORT REVIEW SUMMARY Section
File Review violations?	☐ Yes ☒ No ☐ Not Evaluated	See FILE REVIEW SUMMARY Section

Lead Inspector ID: 304852 Signature:  Date: 5/26/17

Inspection Tracking Information Reviewed by: (1)  (2) (3)

File No. CIWQS Entry Date: 3/29/17 CIWQS Inspection ID: 28029670

CIWQS Coordinator

FACILITY INFORMATION

Riverdale Public Utilities District, WWTF	0.25 mgd
FACILITY DESCRIPTION	DISCHARGE FLOW LIMIT(S)
Secondary non-disinfected	Class I
TREATMENT DESCRIPTION	FACILITY CLASSIFICATION (POTW Only)
Currently unlined evaporation/percolation ponds and land application to ~ 32 acres of pasture	
DISPOSAL DESCRIPTION (e.g., disposal ponds, spreading basins, leachfields, land application area, etc.)	
Screened solids are processed in a comminutor (grinder) and allowed to settle and accumulate in the unlined complete-mix pond. Sludge last removed in 2005 or 2006.	
SLUDGE HANDLING AND STORAGE DESCRIPTION	

WASTEWATER TREATMENT PLANT OPERATOR INFORMATION

Vincent Romero	1	35916
CHIEF PLANT OPERATOR	OPERATOR GRADE	IDENTIFY NUMBER OF OTHER OPERATORS PER GRADE

INSPECTION GIS DATA

GIS Equipment used:				
	MANUFACTURER	MODEL	SERIAL NO.	DATUM

INSPECTION OBSERVATIONS AND FINDINGS

I arrived at the District's WWTF shortly before 2:30 pm. I drove the fenced perimeter of the WWTF and noted standing water in the segment of the old Van Ness Slough and standing water in the ~32-acre land application area. Ms. Madec, Mr. Kemp, Mr. Romero, and Mr. Mauger all arrived shortly thereafter.

The entrance to the WWTF and the headworks is at the southeast corner of the property. We discussed my objectives, which were to familiarize myself with the existing WWTF and the land proposed for the expansion of the WWTF. While the overall appearance of the WWTF is good in that it is free of weeds and well kept, the WWTF is old and in need of replacement. At the entrance to the WWTF, there is an old trickling filter and clarigester that have not been operational since 1985 when the WWTF was last updated (Photograph No.1). The existing WWTF is an aerated lagoon system designed to remove BOD and total suspended solids (TSS). Raw wastewater or influent enters at the headworks where it passes through a comminutor (grinder), prior to being discharged to an unlined complete-mix aerated lagoon (Photograph No. 2). From the complete-mix aerated lagoon, the effluent is discharged to one of six unlined oxidation lagoons, where the effluent percolates into the underlying soil/substrate and evaporates. Effluent in excess of what the ponds can hold is discharged to about 32 acres of adjacent pasture land. A slight sewage odor was noted in the area. Solids are not screened and accumulate in the complete-mix pond. There is limited information on solids disposal and it appears sludge has not been removed since 2003 (see file review).

I walked the perimeter of the complete-mix pond and photographed it and the surrounding "oxidation" ponds (Photographs Nos. 5 and 6). The land application area contained abundant green vegetation and standing water was visible in the land application area and in the segment of the Van Ness Slough that is directly adjacent and west of the western land application area (Photographs Nos. 7 and 8). It was unclear if the standing water in both areas was effluent, rainwater, and/or a combination of both, but it had rained (0.36 inches recorded at Fresno State on 5 March 2017) just two days earlier and this winter season has had higher than average rainfall in the region. The existing pond berms are in generally fair to poor condition, but the southwestern corner of oxidation pond No. 5 had very little freeboard (only about 8 inches). However, the northern pond berm appeared to have about two feet of freeboard. It appears the southern end has sunk or subsided.

The surrounding lands are primarily in agricultural production (Photograph No. 3) and the land directly north contains a confined animal (poultry) operation (Photograph No. 4). There are some rural residences within 5 miles of the WWTF, and the town of Riverdale is about a mile south of the WWTF (Photograph No. 9). There is some oil field activity in the area west of the WWTF as shown in Photograph 10. In summary, the WWTF is old, in disrepair, and is in need of replacement as proposed.

SAMPLING INFORMATION AND OBSERVATIONS

Were samples collected during the inspection? ☐ Yes ☒ No Are sample results included in report? ☐ Yes ☒ No
 Did discharger collect split samples? ☐ Yes ☒ No

SELF-MONITORING REPORT REVIEW SUMMARY

Influent flows are primarily from residential users, but there are some commercial and institutional connections. The existing Waste Discharge Requirements (WDRs), Order 85-252 allow a maximum flow of 0.25 mgd and the average flow into the WWTF in 2016 was 0.21 mgd. The Monitoring and Reporting Program (MRP) doesn't require influent monitoring, but the Discharger analyzed the influent for a select list of constituents on two consecutive days in May 2016 as summarized in the following table. The results are from six samples, three collected on 25 May 2016 and three collected on 26 May 2016. The first number listed is the average and the range of detections is shown below in parentheses.

Influent Results

<u>Constituent</u>	<u>Units¹</u>	<u>Result</u>
Alkalinity as CaCO ₃ and Bicarbonate as CaCO ₃	mg/L	427 (410 - 440)
Biochemical Oxygen Demand	mg/L	105 (71 - 170)
Settleable Solids	mL/L	2.5 (0.9 - 5.3)
Total Kjeldahl Nitrogen	mg/L	45 (42 - 51)
Total Settleable Solids	mg/L	130 (110 - 160)

1. mg/L = milligrams per liter; mL/L = milliliters per liter..

Order 85-252 requires a limited effluent monitoring program for flow, BOD, dissolved oxygen (DO), TSS, and electrical conductivity (EC). The EC results are from 78 weekly samples collected and analyzed from June 2015 through December 2016, while the BOD results are from 18 monthly samples collected and analyzed over the same time period. The results are summarized in the following table.

Effluent Results

<u>Constituent</u>	<u>Units¹</u>	<u>Result</u>
Biochemical Oxygen Demand	mg/L	117 (28 - 170)
Electrical Conductivity	umhos/cm	1,492 (1,016 - 2,010)
12-Month Rolling EC average	umhos/cm	1,443 (1,430 - 1,456)

1. mg/L = milligrams per liter; umhos/cm = micromhos per centimeter.

The current MRP include Discharge Specification B.7 that in part includes BOD limits of 40 mg/L (daily average) and 80 mg/L (maximum). The BOD results of the existing WWTF typically exceed both the 40 mg/L daily average and the maximum of 80 mg/L.

SELF-MONITORING REPORT REVIEW SUMMARY (continued)

Discharge Specification B.10 includes an EC limit of 1,100 micromhos per centimeter (umhos/cm). The elevated EC results are due to the elevated EC levels of the source water that is reported to average about 1,100 umhos/cm. When Order 85-252 was written, source water was from two shallow (~ 500 feet below the ground surface) supply wells that had an EC results of about 600 umhos/cm, but the wells had arsenic in excess of the 10 microgram per liter (ug/L) limit. A deeper well was drilled that met the arsenic limit, but the EC of the source water is about 1,100 umhos/cm. Central Valley Water Board staff will use the Tulare Lake Basin Plan EC limit of 500 umhos/cm plus the EC of the source water or about 1,600 umhos/cm in the new WDRs. The 12-month rolling average of 1,443 umhos/cm meets the limit.

FILE REVIEW SUMMARY

Was a facility file review part of inspection effort?

☒ Yes ☐ No

If not, is facility file review recommended?

☐ Yes ☐ No

The file record (hard copy and electronic) for the Riverdale PUD WWTF is robust and contains several technical reports, notices of violation (NOV), and letters to and from both the Central Valley Water Board and the District.

The CIWQS database contains 272 violations for the District's WWTF dating from February 1997 to April 2016 for WDRs Order 85-252. By far the most common was for exceeding the BOD limits of 40 mg/L daily average and/or the 80 mg/L daily maximum with 108 violations since 1997. The dissolved oxygen results were less than the lower limit of 1.0 mg/L during 51 monitoring events, while the EC exceeded the EC limit (EC of the source water + 500 umhos/cm) during 43 monitoring events since 1997. Incomplete or late reports were submitted 26 times and the flow limit of 0.25 mgd was exceeded 22 times. The discharge occurred off-site on three occasions, sludge was improperly disposed of twice, and there was no documentation for flow meter calibration once.

The CIWQS database contains 10 inspection records dating from March 1994 through March 2017 (this inspection). The file record indicates those inspections led to the issuance of five NOV's dated from June 2001 through June 2006. A June 2001 NOV was issued due to an inspection in April of 2001 that identified that the District was in violation of various portions of WDR 85-252 as follows: Discharge Prohibition A.1 for discharging to a surface water (Van Ness Slough); Discharge Specification B.3 for discharging offsite; Discharge Specification B.7 for BOD being discharged in excess of the 40 mg/L daily average limit; and General Provision A.4 and A.10 of the Standard Provisions for failing to maintain the WWTF. The NOV required the District to submit a Report of Waste Discharge (RWD) addressing additional disposal capacity for the WWTF.

The WWTF was inspected again in February of 2002 due to a complaint by the Consolidated Mosquito Abatement District about standing water in the land application area that was creating nuisance conditions due to the breeding of mosquitoes in the standing water. An NOV was issued in April 2002 for the WWTF being in violation of: Discharge Prohibition A.1 for discharging to a surface water and Discharge Specification B.3 for discharging offsite. The NOV notes that the RWD due in September 2001 was received on 29 March 2002 and would be reviewed.

The WWTF was inspected again on 8 May 2003. A June 2003 NOV was issued to the District for violating the following: Discharge Specification B.4 for exceeding the monthly average flow of 0.25 mgd; Discharge Specification B.5 for disposing of sludge in an unlined pond; and Discharge Specification B.11 for discharging effluent while cattle were present in the land application area. The NOV required the District to submit a technical report evaluating the disposal capacity of the WWTF and a sludge disposal plan by 15 August 2003. The NOV also required the submittal of a RWD by 10 December 2003, indicating the RWD received on 29 March 2002 was incomplete.

FILE REVIEW SUMMARY (continued)

The District was issued another NOV in October of 2003 for continued issues with mosquitoes and the Consolidated Mosquito Abatement District staff and for not submitting the technical report evaluating the disposal capacity of the WWTF and the sludge disposal plan that was required in the June 2003 NOV by August 2003.

In January 2006, the District was issued an NOV for the following violations: Discharge Specification B.5 for unlawful disposing of sludge in an unlined pond; Discharge Specification B.7 for BOD being discharged in excess of the 40 mg/L daily average limit; and Standard Provisions, General Reporting Requirement B.1 for failing to explain violations and proposed remedies in self-monitoring reports (SMRs). The NOV also requests that an amended RWD be submitted.

The District submitted a RWD in January 2006 proposing to add another 40 acres directly west of the existing land application area (current expansion area), build two new disposal ponds, and recycle non-disinfected secondary treated effluent on the adjacent land. On 14 February 2006, The District circulated an Initial Study and Notice of Intent for expansion of the WWTF and the addition of the 40-acre land application area. The project as proposed never happened.

Several letters and reports dated from July 2006 through 7 August 2007 address a proposed groundwater monitoring network for the WWTF. A Groundwater Monitoring Well Installation Workplan was submitted in July 2006 proposing three wells. A 23 August 2006 letter from Central Valley Water Board staff indicates the work plan is incomplete primarily due to there not being an actual upgradient groundwater monitoring well proposed. The District submitted a revised Groundwater Monitoring Well Installation Workplan in December 2006, and Central Valley Water Board staff provided a conditional approval via a letter on 14 February 2007. A 7 August 2007 letter from Provost and Pritchard is in the file summarizing discussions and agreements about the proposed expansion of the WWTF and, in particular the, proposed groundwater monitoring network at the WWTF. Item 2 of the 7 August 2007 letter indicates it was agreed upon to defer the installation of the groundwater monitoring wells until the expansion of the WWTF has begun.

PHOTOGRAPHS



Photograph 1 – Old trickling filter and clarigester (non-operational) at the Riverdale WWTF. View is to the southeast.



Photograph 2 – Looking north from headworks across complete-mix aerated lagoon with the land application area in the background.

PHOTOGRAPHS



Photograph 3 – Adjacent farmland in production directly east of WWTF. The view is to the southeast.



Photograph 4 – Poultry operation (Zacky Farms) north of WWTF.

PHOTOGRAPHS



Photograph 5 - Western edge of Oxidation pond No. 5. Rip rap was placed to minimize erosion of the pond embankment. Southwest corner appears to have subsided and has very little freeboard.



Photograph 6 – Western edge of oxidation pond No. 4 with numerous shore birds present. Lack of rip rap as protection has allowed more erosion of the pond berm.

PHOTOGRAPHS



Photograph 7 – Western part of land application area looking north with standing water visible near center. Poultry operation visible in background.



Photograph 8 – Remnant of the Van Ness slough, just east of the existing WWTF and in the planned expansion area. The slough will be excluded with a buffer zone around it.

PHOTOGRAPHS



Photograph 9 – Community of Riverdale looking east down Mt. Whitney Avenue.



Photograph 10 – Crop land and oil leases about 3 miles west and downgradient of the WWTF.